kök yer eğrileri

1) poles - zeros
2) $\alpha = n - m$
3) centroid : Σ kutup - Σ sıfır

$$G(s)H(s) = \frac{k}{s(s+2)(s+4)}$$

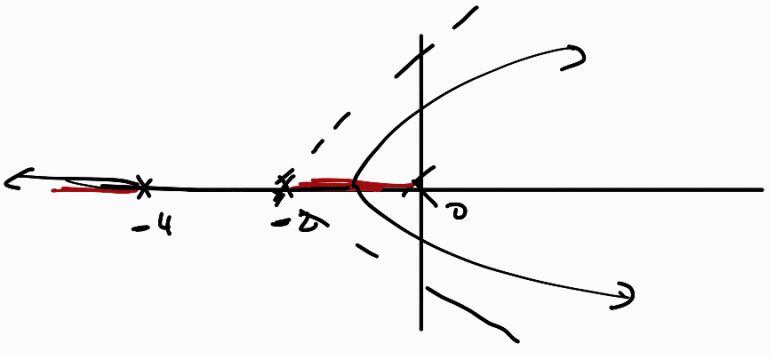
zeros yok $m=0$

pols → $0, -2, -4$ $n=3$

$$\alpha = n - m \quad \swarrow$$

$$\frac{180(2i+1)}{3} = 60, -60, 180$$

$$\sigma_c = \frac{\Sigma \text{ kutup} - \Sigma \text{sıfır}}{\alpha} = \frac{-2-4}{3} = -2$$



$$\Delta = 1 + G(s)H(s)$$

$$\Delta = 1 + \frac{k}{s(s+2)(s+4)}$$

$$\Delta = s^3 + 6s^2 + 8s + k = 0$$

Routh table

s^3	1	8	0
s^2	6	k	
s^1	a		
s^0	k		

Çaprazın bir altına iniyor

$$a = -\frac{\begin{vmatrix} 1 & 8 \\ 6 & k \end{vmatrix}}{6}$$

$$k = -\begin{vmatrix} 6 & k \\ a & 0 \end{vmatrix}$$

$0 < k_c < 48$ ↓ kritik kazanç

$k_c = 48$

$6s^2 + k_c = 0$ ($k_c = 48$)

$s^2 = \sqrt{-8}$

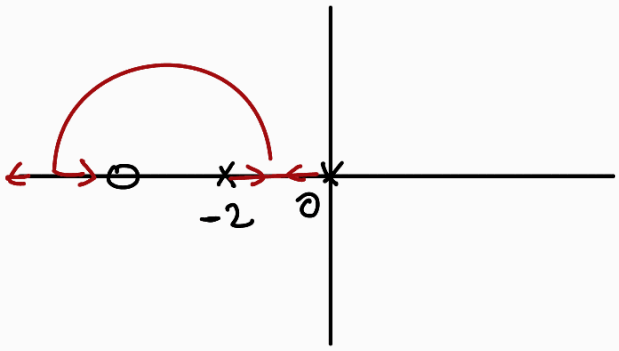
$s = \mp 2\sqrt{2}\,j$ eksenleri kestiği nokta

$$G(s) = \frac{s+4}{s(s+2)} \, k$$

zeros = -4 (m)

poles = 0, -2 (n)

$\sigma = n - m = 1$ //



$$G(s) = \frac{(s+4)\,k}{s(s+1)(s+2)}$$

zeros = -4 $m = 1$

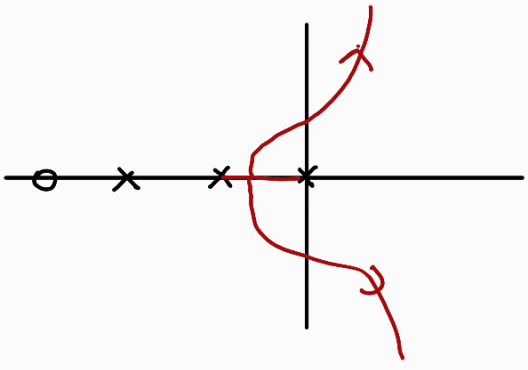
poles = 0, -1, -2 $n = 3$

$\alpha = n - m = 2 \rightarrow \pm 90$

$$\sigma_c = \frac{\sum \text{kutup} - \sum \text{sıfır}}{\alpha}$$

$$= \frac{-4+3}{2}$$

$$= \frac{-1-2+4}{2} = \frac{1}{2}$$



$$G(s) = \frac{(s+4)(s+3)\,k}{s(s+1)(s+2)}$$

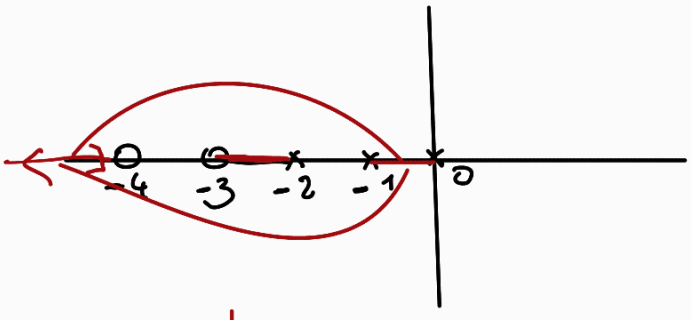
$z = -4, -3 \rightarrow m = 2$

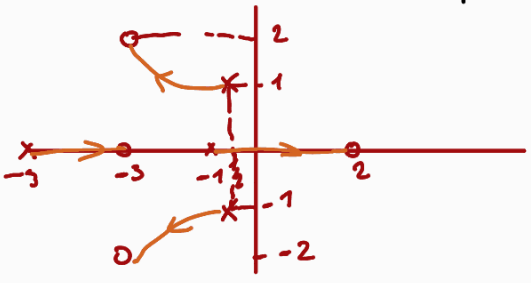
$p = 0, -1, -2 \rightarrow n = 3$

$n - m = 1$

$$-1 - 2 + \frac{4+3}{2}$$

$$= 4$$





$$\frac{\text{zeroes}}{\text{poles}} = \frac{(s-2)(s+3)\,(s+3+2j)(s+3-2j)}{(s+5)(s+1)\,(s+\frac{1}{2}-j)\,(s+\frac{1}{2}+j)}$$

$n - m = 0$

	step	ramp	
0	$\frac{1}{1+k_p}$		
1		$\frac{1}{k_v}$	
2			$\frac{1}{k_a}$

$$k_p = \lim_{s \to 0} G(s)$$

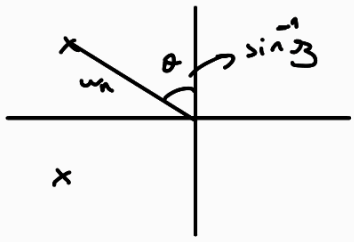
$$= \frac{-2 \cdot 3\,(3+2j)(3-2j)}{5 \cdot 1\,(\frac{1}{2}-j)(\frac{1}{2}+j)}$$

$$= -4{,}16$$

$$\frac{1}{1+k_p} \rightarrow -0{,}316$$

6-3 = 3

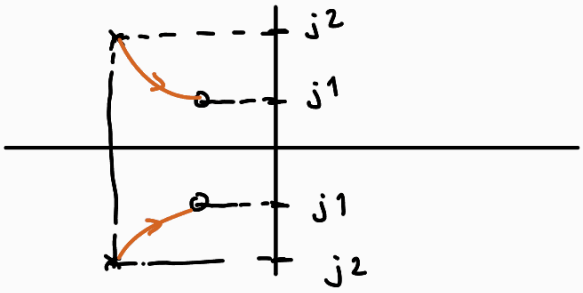
-60 +60 +180



$$G(s)\,H(s) = \frac{k\,(s+1-j)\,(s+1+j)}{(s+2+2j)\,(s+2-2j)}$$

zeros: $-1+j$, $-1-j$

poles = $-2j$, $-2+2j$





$$\underbrace{(\Phi_1 + \Phi_2)}_{\text{sıfırlar}} - \underbrace{(\Theta_1 + \Theta_2)}_{\text{poles}} = \mp 180$$

$$\Theta_1 = \mp 180 + \underbrace{\Theta_2}_{\downarrow 90} - \underbrace{\Phi_1}_{\sim 135^\circ} - \Phi_2$$

Tekrar

$M_p = \%25$ $\quad t_p = 2sn$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} = 0,25$$

$$\frac{-\pi\zeta}{\sqrt{1-\zeta^2}} = \ln 0,25$$

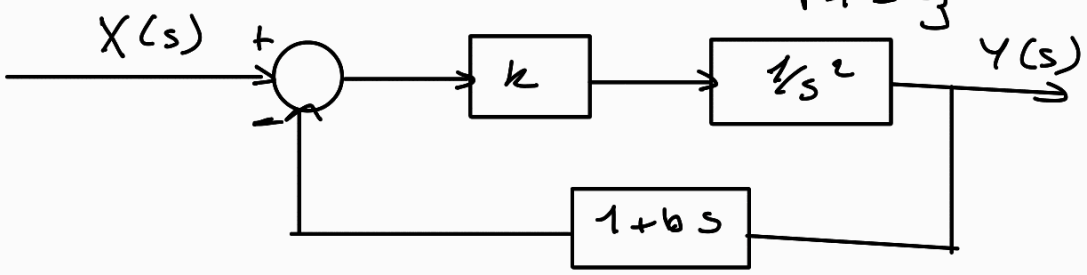
$$\frac{\zeta}{\sqrt{1-\zeta^2}} = 0,441$$

$$\frac{\zeta^2}{1-\zeta^2} = 0,194 \qquad \zeta^2(1+0,194) = 1$$

$$\zeta = 0,40$$

$t_p = \frac{\pi}{\omega_d}$ (2s) $\omega_d = \frac{\pi}{2}$ $\frac{\pi}{2} = \omega_n\sqrt{1-\zeta^2}$

$$\omega_n = \frac{\frac{\pi}{2}}{\sqrt{1-\zeta^2}} = 1,71$$



$$G(s) = \frac{k\,\frac{1}{s^2}}{1 + \frac{k(1+bs)}{s^2}}$$

$$= \frac{k}{s^2 + k + kbs}$$

$$= \frac{k}{s^2 + kbs + k} \qquad \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$k = \omega_n^2$ $k = \omega_n^2 = 2,92$

$kb = 2\zeta\omega_n$

$$b = \frac{2\zeta\omega_n}{\omega_n^2} = 0,46$$

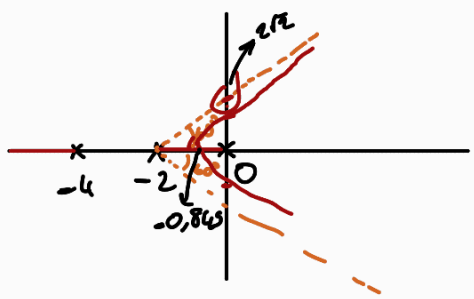
2)

$$G(s)H(s) = \frac{k}{s(s+2)(s+4)}$$

zeroes ∄

poles $p_1 = 0$ $p_2 = -2$ $p_3 = -4$ $n = 3$

$a = n - m = 3 - 0 = 3$



$$\text{centroid} = \frac{\Sigma \text{poles} - \Sigma \text{zeros}}{a}$$

$$= \frac{-2-4}{a} = \frac{-6}{a} = -2$$

$$\text{Asymptotes angles} = \pm\frac{180(2i+1)}{3} \quad i = 0, \ldots$$

$$= 60, 180, \underset{300}{-60}$$

$$\Delta = \frac{s(s+2)(s+4)+k}{s(s+2)(s+4)} = 0$$

$$s(s+2)(s+4)+k = 0$$

$$k = -\underbrace{s(s+2)(s+4)}_{s^3+6s^2+8s}$$

$$\underbrace{s^2+2s}_{s^3+2s^2+4s^2+8s}$$

$$\frac{dk}{ds} = -(3s^2+12s+8)$$

kökleri $(3s^2+12s+8)$

$s_1 = -0{,}845$, $s_2 = -3{,}154$ ✗ non acceptable

↓ departure point

$k = 3{,}07$

k range

$$s^3+6s^2+8s+k = 0$$

s^3	1	8	0
s^2	6	k	
s^1	$\frac{48-k}{6}$	0	
s^0	k		

$48-k > 0$

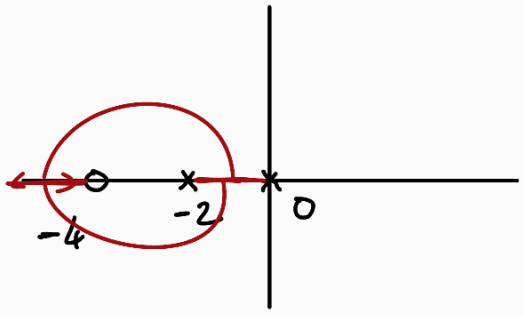
$0 < k < 48$

$k = 48$

$s^3+6s^2+8s+48 = 0$

$s_1 = -6$ $s_2 = 2\sqrt{2}\,j$ $s_3 = -2\sqrt{2}\,j$

$$G(s) = \frac{s+4}{s(s+2)}\, k$$



zeroes

$s = -4$

$p_1 = 0 \quad p_2 = -2$

$\alpha = n - m = 1$

∓ 180 (kök)

-1 e kutup ekleyelim

$$G(s) = \frac{(s+4)\, k}{s(s+1)(s+2)}$$

zeroes

$z_1 = -4$

$p_1 = 0 \quad p_2 = -1 \quad p_3 = -2$

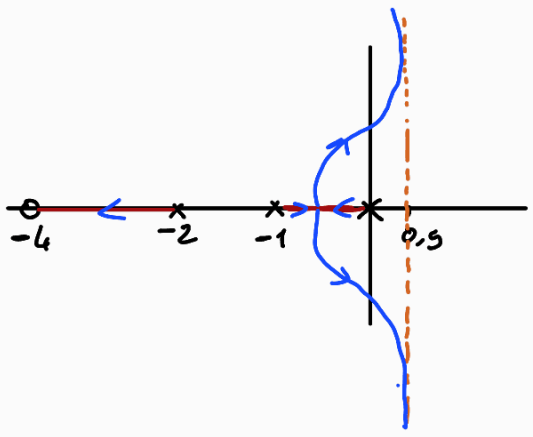
$n - m = 3 - 1 = 2$

$$\text{centroid} = \frac{-1-2+4}{2} = \frac{1}{2} = 0{,}5$$

$$\text{angles} \quad \frac{180(2i+1)}{2}$$

$+90$

-90



$$s(s+1)(s+2) + (s+4)k = 0$$

$$(s+4)k = -((s^2+s)(s+2))$$

$$k = \frac{-(s^3+3s^2+2s)}{(s+4)}$$

$$\frac{dk}{ds} = \frac{-(3s^2+6s+2)(s+4) + (s^3+3s^2+2s)(1)}{(s+4)^2} = 0$$

$$-3s^3 - 12s^2 - 6s^2 - 24s - 2s - 8 + s^3 + 3s^2 + 2s = 0$$

$$-2s^3 - 21s^2 - 24s - 8 = 0$$

$$2s^3 + 21s^2 + 24s + 8 = 0$$

$s_1 = 5{,}42$ (X) $\quad s_2 = -0{,}45$ $\quad s_3 = -1{,}62$ (a)

$s(s+1)(s+2)+(s+4)k=0$

$(s+4)k=-((s^2+3)(s+2))$

$k=-\frac{(s^3+3s^2+2s)}{(s+4)}$

$s^3+3s^2+2s+ks+4k=0$

$s^3+3s^2+(2+k)s+4k=0$

s^3	1	$2+k$
s^2	3	$4k$
s^1	$\frac{6+3k-4k}{3}$	
s^0	0	

$6-k>0$

$k<6$ → critical gain

$0<k<6$ //

$s^3+3s^2+8s+24=0$

$s_1=-3$

$s_2=2{,}82j$

$s_3=-2{,}82j$

⇒ 3 zero ekle

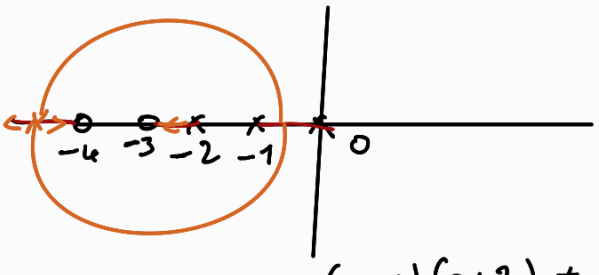
$G(s)=\frac{(s+4)(s+3)k}{s(s+1)(s+2)}$

$z=-4, \rightarrow m=2$

$p=0,-1,-2, n=3$

$a=n-m=1$

angle $=\pm 180$

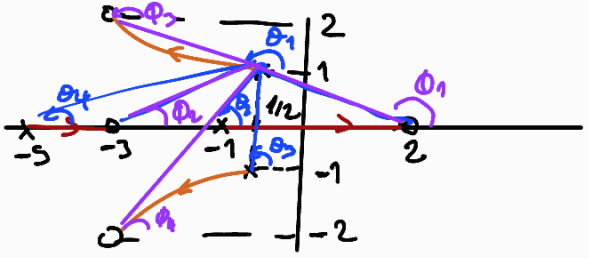


$s(s+1)(s+2)+\underbrace{(s+4)(s+3)}_{s^2+3s+4s+12 \;\; s^2+7s+12}k=0$

$s^3+3s^2+2s+ks^2+k7s+12k=0$

$k(s^2+7s+12)=-$

$s^3+(3+k)s^2+(2+2k)s+12k=0$



$n-m=0$ 0 sonsuz

$$G(s) = \frac{(s-2)(s+3)(s+3+2j)(s+3-2j)}{(s+5)(s+1)(s+\frac{1}{2}+j)(s+\frac{1}{2}-j)}$$

$$E(s) = \frac{R(s)}{1+G(s)} \to \frac{1}{s}$$

$$\lim_{s\to 0} sE(s) = \frac{1}{1+\underbrace{\lim_{s\to 0} G(s)}_{-12,48}} = -0,0871$$

$$E_{ss} = -0,0871$$

Skill Assesmen Ex 8,4

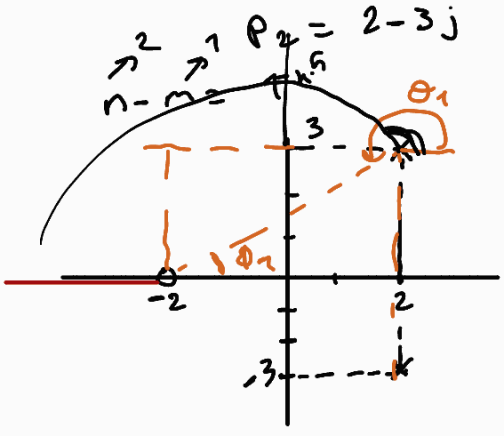
Unity feed back system

$$G(s) = \frac{k(s+2)}{(s^2-4s+13)}$$

a) $z_1 = -2$

$p_1 = 2+3j$

$p_2 = 2-3j$



$$\Phi_1 - \theta_1 - \theta_2 = 180$$

$$\underbrace{\arctan\left(\frac{3}{4}\right)}_{36,83} - \theta_1 - 90^\circ = 180^\circ$$

$$\theta_1 = 36,83 - 90 - 180$$

$$= -233$$

$$\Delta = 1 + G(s) = 0$$

$$s^2 - 4s + 13 + k(s+2) = 0$$

$$k = -\frac{(s^2-4s+13)}{(s+2)}$$

$$\frac{dk}{ds} = \frac{-(2s-4)(s+2) + (s^2-4s+13)}{(s+2)^2} = 0$$

$$-(2s^2+4s-4s-8)+(s^2-4s+13)=0$$

$$-2s^2+8+s^2-4s+13=0$$

$$-s^2-4s+5=0$$

$$\Rightarrow s^2+4s-21=0$$

$s_1=3$

$s_2=-7$

break in point

$$s^2-4s+13+k(s+2)=0$$

$$s^2+(-4+k)s+13+2k=0$$

s^2	1	$13+2k$	
s^1	$k-4$	0	0
s^0	$13+2k$	0	

$k-4>0$

$k>4$

$k-4=0$

$k=4$ olsun

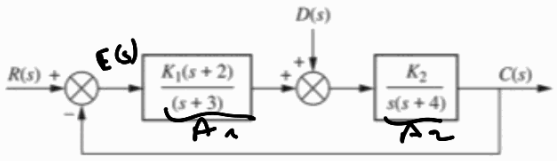
$$s^2+13+\underbrace{2k}_{8}=0$$

$$s^2=-21$$

$$s=\mp j\sqrt{21}$$

2- In the given system, determine the two gains K_1 and K_2 so that both following conditions are satisfied,

- Steady-state error due to a unit step disturbance is -0.00001
- Steady-state error due to a unit step input is 0.002



$E(s) = R(s) - C(s) \rightarrow C = R - E$

$C = (E A_1 + D) A_2$

$R - E = E A_1 A_2 + D A_2$

$R - D A_2 = E (1 + A_1 A_2)$

$$E = \frac{R}{1 + A_1 A_2} - \frac{D A_2}{1 + A_1 A_2}$$

if $R = 0 \quad D \neq 0 \quad D = \frac{1}{s}$

$$E = -\frac{D A_2}{1 + A_1 A_2}$$

$$E_{ss} = \lim_{s \to 0} sE(s) = \frac{-A_2}{1 + A_1 A_2}$$

$$= \lim_{s \to 0} \frac{-\frac{k_2}{s(s+4)}}{1 + \frac{k_2}{s(s+4)} \frac{k_1(s+2)}{(s+3)}}$$

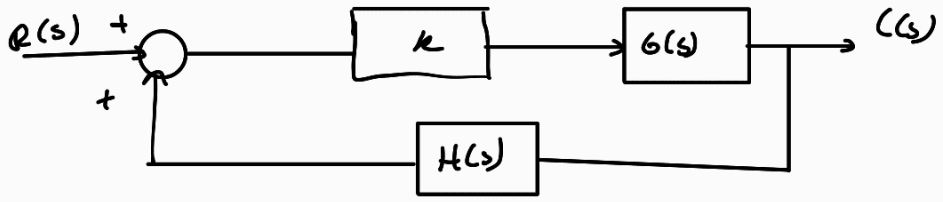
$$\rightarrow 0{,}0001 = \lim \frac{k_2(s+3)}{s(s+3)(s+4) + k_1 k_2 (s+2)} = \frac{3k_2}{2k_1 k_2} = \frac{3}{2k_1}$$

$k_1 = 150{,}000$

b) $R(s) = \frac{1}{s} \quad D(s) = 0$ is 0,002

$$0{,}002 = \lim_{s \to 0} \frac{1}{1 + G_1 G_2} = \lim_{s \to 0} \frac{s(s+3)(s+4)}{s(s+3)(s+4) + k_1 k_2 (s+2)} \neq 0$$

Positive Feedback Architecture



$$\frac{C(s)}{R(s)} = \frac{kG(s)}{1 - kG(s)H(s)}$$

$$1 - kG(s)H(s) = 0$$

$$|G(s)H(s)| = \frac{1}{k}$$

Arrival angle

$$360 = \text{zeros angle} - \text{poles angle}$$

(3) Assymptotic angle

$$\pm \frac{360}{n-m}$$

---

zero: lead ⟶ sola kayar root efendi daha stabil, hızlı cevap accuracy kötü

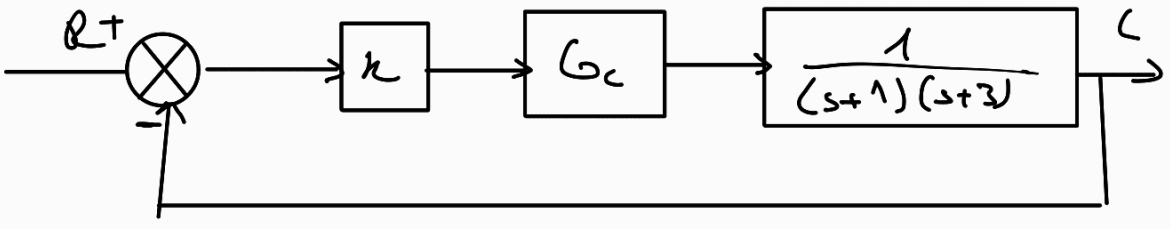
poles: lag

$$G_c = k_c \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}$$

$0 < \alpha < 1$: lead

$\alpha >$: lag

Ex



Dominant pole $= -3 + 2j$

steady steady error: 10%

27. For the unity feedback system shown in Figure P8.3, where

$$G(s) = \frac{K(s^2 - 2s + 2)}{(s+2)(s+4)(s+5)(s+6)}$$

$(1+j)\ (1-j)$

do the following: [Section: 8.7]

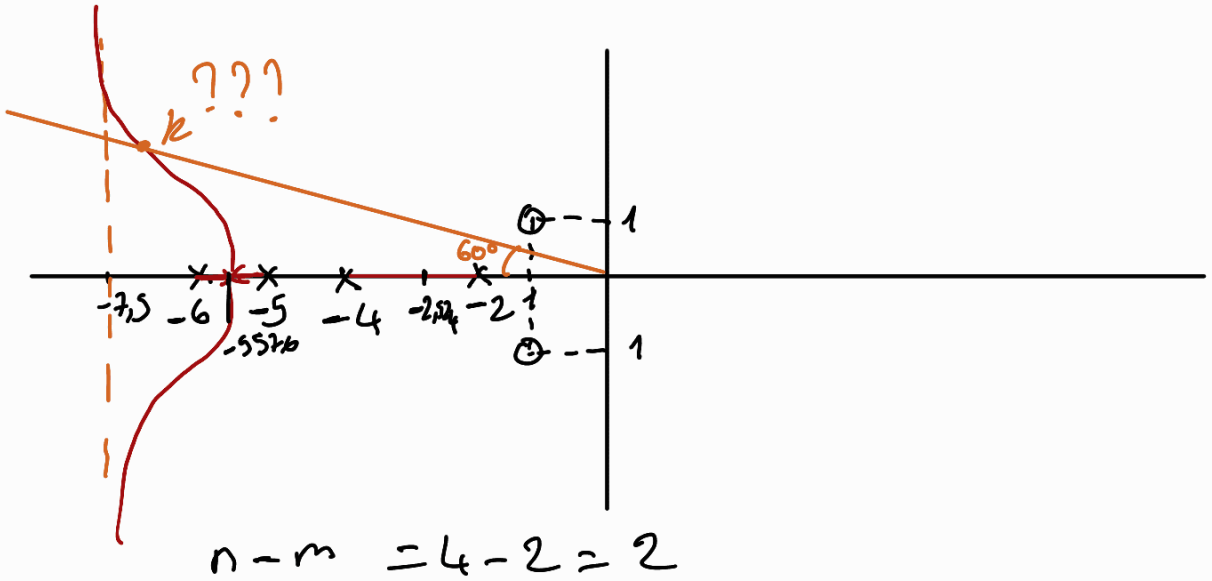
a. Sketch the root locus.

b. Find the asymptotes.

c. Find the range of gain, K, that makes the system stable.

d. Find the breakaway points.

e. Find the value of K that yields a closed-loop step response with 25% overshoot.



$n - m = 4 - 2 = 2$

$$\sigma_c = \frac{\Sigma \text{r-p. poles} - \Sigma \text{r-p. zeroes}}{2}$$

$$\sigma_c = \frac{-6-5-4-2+1+1}{2} = \frac{-15}{2} = -7{,}5$$

$$\phi = \frac{(2i+1)}{2} = \pm 90$$

$$\Delta(s) = 1 + \frac{K(s^2 - 2s + 2)}{(s+2)(s+4)(s+5)(s+6)}$$